



Estimation of voter transitions and the ecological fallacy

Antonio Forcina¹ · Davide Pellegrino²

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Abstract

This paper attempts an investigation into the features of ecological fallacy in the context of estimation of voter transitions between two elections. After reviewing some theoretical findings from a statistical point of view, we discuss two tools for checking whether bias is present: (1) fitting models with covariates; (2) comparing the standard errors of transition probabilities computed under ideal conditions against those based on bootstrap methods. Concerning the effect of covariates, we describe two different data generating mechanisms, depending on whether voting decisions are affected by variables measured at the (1) aggregate or (2) the individual level. By theoretical arguments and empirical evidence, we show that, while modelling the effect of covariates removes bias in the first case, it may fail in the second because only aggregate level covariates are usually available. Our investigation relies on the analysis of real and artificial data sets: the latter are obtained by a computer software which mimics voting behaviour and is such that, artificial electoral data with designed size and direction of ecological bias can be produced. An application to a recent election in the city of Turin is also used to illustrate our methodology and findings.

Keywords Ecological bias · Effects of covariates · Voting transitions · Bootstrap

1 Introduction

Estimates of voter transitions from one election to the next, may be displayed as a two-way contingency table with the voting options in the earlier election listed by row and those of the new election by column; the values along each row indicate how, those who choose that option in the previous election, split among the options available in the new election. This table, if available, provides several important insights on voting behaviour, for instance:

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✉ Antonio Forcina
forcinarosara@gmail.com

Davide Pellegrino
davide.pellegrino@unito.it

¹ Dipartimento di Economia, via Pascoli, 06100 Perugia, Italy

² Interuniversity Department of Regional and Urban Studies and Planning (DIST), Università degli Studi di Torino: Politecnico di Torino, viale Mattioli 39, 10125 Turin, Italy

- the proportion of faithful voters (that is those who voted the same party in both occasions);
- the size of protest voters who moved towards minor parties or to abstention;
- the ability of an emerging party to recover voters from abstention;
- the size and the direction of strategic voting (Herrmann and Pappi 2008), when the supporters of candidates not admitted to the runoffs, have to decide whether to abstain or support one of the competing candidates.

Due to the special nature of electoral data, the only information available are the so called *aggregate data*, that is the frequency distributions of voters by local unit and voting options, separately, in each election. Direct estimates of voter transitions would require access to *individual data*, that is the frequency distribution of voters according to the pair of voting options selected in the two elections in each local unit or, at least, the *condensed* versions of individual data, when the joint distribution is known only for the area under investigation as a whole.

Voter transitions could be estimated by ecological inference or exit polls. However, as discussed, for instance, in Russo (2014), estimates from exit polls may be seriously biased due to non response, voters not telling the truth or not remembering their earlier decision. There are additional limitations with exit polls:

- it may be difficult to sample from abstainers as they do not show up at the polls;
- while a sample size of a few thousands may provide accurate estimates of the proportion of voters for each party in a given election, a much larger sample size is necessary to estimate how the voters of a small party split among the options available in the new election;
- often, voter transitions have strong local features which would cancel out when sampled on a whole region or State.

A comparison between estimates from exit polls and ecological inference was presented by Liu (2007) in a specific context where individual data were available. Condensed individual data may be available in the context of simultaneous elections like in the case examined by Plescia and De Sio (2018).

Due to the limitations of exit polls, in most situations ecological inference is the only feasible approach for the estimation of voter transitions. Thus it may be useful to study in depth, in the context of electoral data, under which conditions these estimates may be biased, a phenomenon known as ecological fallacy. A brief history of the subject starts with the seminal paper by Robinson (1950); substantial contributions were also provided by Firebaugh (1978) and, more recently (Wakefield 2004). By elaborating on the last two papers, Gnaldi et al. (2018) provide a more explicit characterization of the necessary and sufficient conditions for ecological inference to provide unbiased estimates; these results will be the starting point of the present paper which relies on the analysis of real and artificial electoral data generated according to a well defined model of voting behaviour presented in Sect. 4.

We also discuss how to detect the presence of ecological bias when individual data are not available and concentrate on two approaches: the fitting of models where transition probabilities are allowed to depend on covariates and bootstrap methods, see for instance Efron and Tibshirani (1994).

The paper is organized as follows. In Sect. 2 we define the setting and summarize the basic results concerning ecological fallacy and diagnostic tools to detect its presence. In

Sect. 3 we investigate the possibility of correcting bias by modelling the effect of aggregate and individual level covariates. In Sect. 4 we describe an algorithm for generating artificial electoral data according to the modified Brown and Payne model in Forcina et al. (2012) and analyse three data sets affected by different types of ecological fallacy, including an instance where voter decisions are assumed to depend on individual level covariates. In Sect. 5 we present an application to the election for Mayor in the city of Turin in 2016 by fitting models with covariates to reduce bias, a procedure which is validated by a bootstrap simulation.

2 The ecological fallacy revisited

Suppose that we are interested in voting transitions within a relatively small area like a town made of K polling stations. Let Y denote the choice of a voter among the C options available in a given election and X the corresponding choice among the R options that were available in a previous election. Let p_{uij} denote the proportion of voters who chose $Y = j$ among those who voted $X = i$ in the previous election in polling station u . Let also π_{ij} denote the probability that an eligible voter chosen at random among those who voted $X = i$ will chose $Y = j$; these quantities are the main object of ecological inference. They will be called *voter transitions* or *transition probabilities* though, when $i = j$, they measure the faithfulness of voters of party i ; note that they sum to 1 by row.

Let also y_{uj} and x_{ui} denote, respectively, the aggregate proportions of voters for party j in the new election and for party i in the previous election in polling station u . By simple algebra, as for instance in Gnaldi et al. (2018), it can be shown that the usual ecological regression may be written as

$$y_{uj} = \sum_i^R x_{ui} \pi_{ij} + \epsilon_{uj}, \text{ where } \epsilon_{uj} = \sum_i^R x_{ui} (p_{uij} - \pi_{ij}); \quad (1)$$

a closer investigation of the nature of the error term ϵ_{uj} indicates that the condition for unbiased estimates by any methods based on linear or non linear regression on the marginal proportions x_{ui} is that these are uncorrelated with the p_{uij} , the proportion of voter transitions within each polling stations. The amount of bias produced by correlations between the p_{uij} and the x_{ui} depends on the signs and the strength of the correlations; as explained by Wakefield (2004), section 3.3, in the 2×2 case the largest biases are to be expected when correlations are all in the same direction. Predicting size and direction of bias in general is very complicated, some insights are provided in the examples presented in Sect. 4.

In short, there will be ecological bias whenever the variations across polling stations of any of the p_{uij} is correlated with any of the marginal proportions x_{ui} ; this is the *x-bar* rule in Firebaugh (1978) terminology. Note that the more stringent condition (Goodman 1953, pag. 664) according to which the expected values of the p_{uij} must be constant across local units is sufficient but not necessary to prevent ecological bias. The condition stated in Goodman (1959, pag. 624) is more accurate and, essentially, equivalent to (1) above which says that, we are exposed to ecological bias unless voting transitions and marginal proportions are uncorrelated. Associations between these two sets of quantities may be intrinsic, like when, for instance, the proportion of faithful voters of a given party is greater in those polling stations where the party was stronger in the previous election. Association may also be induced when certain entries of the table of voter transitions are correlated with an

external variable which, in turn, is correlated with the marginal proportions; an example of this phenomenon is outlined in Sect. 4.2.

We may also say that ecological bias is present whenever the direction and strength of the associations at the aggregate level differ from those at the individual level. The notion of *individualistic fallacy*, discussed by Welzel and Inglehart (2007), denotes the attitude of those who deduce “the non-validity of an existing relation at the aggregate level from the non-existence of the same relation at the individual level”; for a more in depth discussion and additional references, see Russo (2017). This is an interesting point of view and a subject of debate among political scientists; for a different point of view see for instance Seligson (2002). In the context of vote transitions, the connection between individual voting decisions and certain variables measured at the aggregate level are relevant when considering the effect of covariates examined in Sect. 3.

2.1 Robinson's data

A rather complex set of multilevel models were fitted by Subramanian et al. (2009) to a slightly extended version of the data used by Robinson (1950); though these data are not about voter transitions, they may be useful for clarifying the nature of the ecological fallacy. To test the condition for ecological fallacy in this context, note that the proportion of residents belonging to the different ethnic groups (Whites born in US, Whites not born in US and Blacks) are the analog of our x_{ui} , the proportion of votes obtained by each party in the previous election, while the proportion of illiterate within each ethnic group, are the analog of our p_{uij} ; because $C = 2$, $p_{ui2} = 1 - p_{ui1}$ and we may restrict attention to p_{ui1} . The issue is whether the variations by State of p_{ui1} , the proportions of illiterate, are correlated with those of the x_{ui} .

An informal assessment may be obtained by examining the quantile regression plots of Fig. 1; these were obtained by classifying States in 10 groups according to the deciles

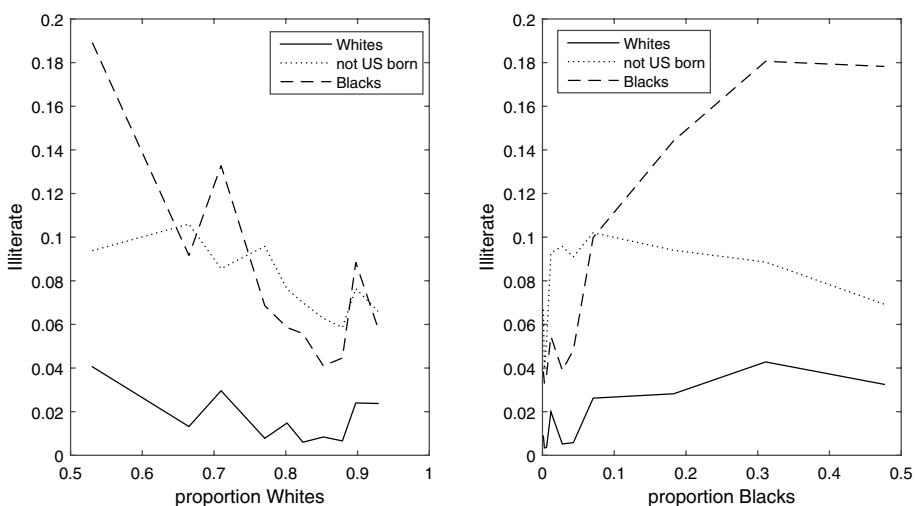


Fig. 1 Quantile regression of the proportion of illiterate by race against the proportion of Whites US born (left panel) and the proportion of Blacks (right panel)

of the explanatory variables (for instance the proportion of Black residents). Then, within each group of States, we plot the average proportion of illiterates by race against the average proportion of Blacks (or Whites).

It emerges clearly that the proportion of illiterate decreases even among Blacks when the proportion of Whites increases; the relation is reversed, though with a different shape, when the proportion of Blacks increases. Robinson's data were also reanalysed by Firebaugh (1978, Section IV) who argued that region (South/rest of the Country) was the very determinant of illiteracy. In this example the proportion of Blacks is positively correlated with the proportion of illiterates within Blacks and Whites US born.

2.2 Tools for detecting the ecological fallacy

In the unlikely event that the full individual data were available, one could simply compute the p_{uij} proportions and plot these as functions of the marginal proportions x_{ui} ; for instance, for a given i we might choose a set of j s of interest. A quantile regression as in Fig. 1 may be used for a visual inspection of the overall direction of correlations.

When only the condensed individual data are available, as in Plescia and De Sio (2018), for each cell in the table of voter transitions, one can compute the errors in the estimates from ecological inference relative to the true values. These errors, as in any estimation problem, will be the sum of bias and random variation, thus, additional information or specific assumptions are required to interpret these results.

From the aggregated data alone we cannot even know which estimates are closer to the truth. A set of indirect diagnostic tools for checking whether conditions for unbiased estimates are satisfied were discussed by Loewen and Grofman (1989) but criticized by Klein et al. (1991). In this paper we propose two diagnostic tools more directly related to the nature of ecological fallacy: (1) fitting models with covariates, (2) comparison of standard errors estimated from theory with the corresponding bootstraps estimates. Point (1) will be the subject of next section; while (2) is described below. An application of the bootstrap will be presented in Sect. 5 and a small simulation study, based on artificial electoral data, will be presented in Sect. 4.

The bootstrap is a statistical method for computing non parametric estimates of standard errors (Efron and Tibshirani 1994); to apply this to the context of ecological inference, one has to select, at random, several subsets of local units and compute the estimates of voter transitions by the chosen method within each subset. Intuitively, if variations of transition probabilities across polling stations are not purely random, the estimates obtained from different samples are likely to exhibit an amount of variability larger than expected from theory. The method can be computationally demanding because the chosen method of ecological inference has to be applied, say, 300 times. In the end, 300 different estimates of each entry of the table of voter transitions will be available so that uncertainty in each estimate can be assessed without the need to make any assumption.

The bootstrap estimates of standard errors may be compared with the corresponding estimates derived from theory under the assumption that the model is correctly specified. In particular, for the Brown and Payne model, the fitting algorithm provides an estimate of the covariance matrix of the parameters on the logit scale derived from the expected information matrix. Because transition probabilities are a non linear transformation of the underlying parameters, the Delta method (Wooldridge 2010, pag. 44, 45) may be used to compute the variance of transition probabilities.

The following testing procedure may be used to decide whether the bootstrap standard errors are significantly larger than the theoretical ones. An $R \times C$ table of transition probabilities has RC entries; under the null hypothesis of no bias we expect that each difference between the bootstrap and the theoretical standard error may be positive or negative with equal probabilities (though transitions sum to 1 by row, the corresponding standard errors are unconstrained). Thus the symmetric binomial distribution may be used to compute a significance level; in practice, we need to count the number of time that the bootstrap standard error is larger than the corresponding theoretical value and compute a significance level.

An application to artificial data will be presented in Sect. 4; the method will also be used in Sect. 5 to provide evidence that a certain amount of bias is removed by the model with covariates.

3 Ecological fallacy and covariates

Several methods of ecological inference allow to model the effect of covariates, measured at the level of polling stations, on transition probabilities, for instance Brown and Payne (1986) and Rosen et al. (2001); this possibility is also mentioned as an extension by Greiner and Quinn (2009). However, as far as we know, the fact that we can restrict attention to the covariates which are correlated with the x_{ui} (the proportion of voters for the different options available in the previous election) has not been noted explicitly.

Individual data from the 2002 New Orleans mayoral election were used by Liu (2007) to show that, by modelling the dependence of transition probabilities on appropriate covariates, one can reduce substantially the ecological bias. This is not surprising because, a correctly specified statistical model leads to consistent estimates. An example based on computer generated electoral data that confirms this expectation will be given in Sect. 4.

The assumption that the logits of transition probabilities can be a linear function of covariates measured at the level of polling station is stated by Rosen et al. (2001) in their equation (2); an identical assumption is formulated by Brown and Payne (1986) in their equation (2.3). Both formulations can be rewritten as

$$\log \frac{\pi_{uij}}{\pi_{uiC}} = \alpha_{ij} + \beta'_{ij} z_{uij}, \quad (2)$$

where α_{ij} is an intercept parameter, π_{uij} is the transition probability from i to j in local unit u , z_{uij} is a vector of covariates affecting cell (ij) and β_{ij} is a vector of regression coefficients acting on the logit scale.

Covariates measured at the level of polling stations which might affect voting decisions are, for instance, the proportion of people aged over 65, the proportion of unemployed or the proportion of immigrants. The idea behind the method is that, if voter transitions are not constant and their variation across polling stations are determined by certain covariates, which are correlated with the marginal proportions as in Eq. 2, then the induced ecological bias will be removed.

A detailed discussion of social mechanisms by which individual behaviours may be affected by average features of a local unit (polling station) are discussed in depth by Firebaugh (1978, pp 564–568). A specific examination concerning the way that the environment voters live in can affect their decisions is contained in Johnston and Pattie (2006), in particular the chapter: “*Talking together, voting together*”.

In addition to covariates measured at the aggregate level, which may affect all voters of a certain category living in the same local unit, voters may also decide on the basis of their personal condition (like, for instance, being unemployed, elder, in poverty), rather than on the value of the same variables aggregated at the level of their polling station. Covariates of this type were called *micro properties* by Firebaugh (1978).

If voters' decisions were determined by the individual value of certain covariates, using Eq. (2) would be equivalent to use a misspecified model which may fail to correct bias. Unfortunately, as explained by Gnaldi et al. (2018), there is no known alternative to (2) to account for the effect of individual level covariates. An artificial data set where aggregate level covariates fail to correct the bias induced by individual level covariates is presented in Sect. 4.

4 Ecological fallacy in artificial electoral data

4.1 Outline of the data generating mechanism

In the simulations below we use a data generating mechanism based on the revised version of the Brown and Payne model described by Forcina et al. (2012) where the interested reader can find a detailed description of the statistical model. An outline of the procedure derived from this model is given below; the MATLAB functions that implement the procedure with instructions on their usage is included as supplementary material.

Allocation of voters among the options available in a given election is achieved by sampling from a Multinomial-Dirichlet (MD in the following) distribution (see, Ng et al. 2011, Chap. 6); this is equivalent to the drawing of balls of different colours from a set of urns (clusters of similar voters) whose composition is allowed to vary at random to mimic the fact that voting behaviour may vary at random even within polling stations according to specific local features. The model is determined by the set of average allocation probabilities and a coefficient that determines the amount of random variation across clusters. The data generating mechanism may be summarised as follows:

1. Voters within each polling station are assigned to the options available in the first election by sampling from an MD model with equal probabilities.
2. The logits corresponding to each row in the table of transition probabilities within each polling station are computed as in Eq. (2) where the z_{uih} are made equal to the x_{ui} .
3. The procedure to generate the distribution of voters in the second election uses the following steps; (1) for each u (polling station) and i (option voted in election 1) the n_{ui} voters are split at random in c clusters; (2) within each cluster, voters are allocated to the options available in the second election by sampling from an MD model where the transition probabilities are determined by Eq. (2) and have a cluster specific random effect.
4. By aggregating voters from different clusters within the same polling station, the individual level distribution of voters in each polling station is computed.

To generate data where voting decisions depend on individual level covariates, a more complex procedure is required, as outlined below; to be specific, assume that the individual level covariates is whether a given voter is unemployed or not, then:

1. Within each polling station, split voters between employed and unemployed by sampling from an MD model.
2. Split, separately, employed and unemployed among the options available in the first election by sampling from an MD model but according to probability distributions which are different for employed and unemployed.
3. Apply step 3 above to split voters with a given occupational status, who chose a given option in the previous election, among the options available in the new election.
4. At the end, for each polling station, the individual data will consist of the frequency distribution of voters classified by employment status, option voted in the first election and option voted in the second election.

The first generating mechanism is used to produce data affected by ecological bias in different directions and to show that the three main methods of ecological inference produce biased estimates unless a proper model for covariates is used. The second generating mechanism is used to show that, under certain conditions, when voting decision depend on individual level covariates, modelling the effect of aggregate level covariates may fail to correct ecological bias.

4.2 Ecological bias and covariates

The artificial data will be analysed with three methods of ecological inference: the Goodman linear regression, the frequentist version of the Multinomial-Dirichlet method by Rosen et al. (2001) and the modified version of the Brown and Payne model in Forcina et al. (2012). The objective here is not to compare the three different methods but to show that bias affects all methods approximately in the same way and that the bias disappears only if a correctly specified model for the effect of covariates is used. To minimize random variation and thus allow the bias to emerge more clearly, the three artificial data contain 20,000 polling stations.

We first analyse two artificial data sets generated by assuming that: (1) there are three options available in each election; (2) on average, 60% of voters are faithful to the option voted in the first election and the remaining 40% split equally between the other two options; (3) the faithfulness to the option h voted in election 1 is positively (negatively) correlated with the marginal proportion x_{1h} . The results displayed in Table 1 indicate that all ecological inference methods overestimate the voter transitions along the main diagonal when correlations are positive and underestimate the same transitions when correlations are negative. When a correct model for the effect of covariates is used, the average absolute error drops below 0.0032 for both the Brown and Payne and the King methods. These results are in accordance with the fact that both methods may be seen as instances of M-estimation in non linear models which can be shown to be consistent (Wooldridge 2010, 12.2); it follows that the average absolute error may be attributed to random variation.

As an additional check, we analysed an artificial dataset within the same setting described above, where correlations between voter transitions and marginal proportions are null so that ecological bias is absent. In this case the three methods (Goodman, King and Brown and Payne) produce estimates with an average absolute error below 0.0014.

The objective of the last artificial data set is to present an instance of a setting where voter decisions depend on individual level covariates and modelling the effect of aggregated level covariates fails to correct bias. To keep the context as simple as possible, we assume that there are only two voting options in both elections, leading to a 2×2 table

Table 1 Estimation errors in voter transitions without covariates, relative to individual data

Estimation method									
G				K			BP		
U	V	Z		U	V	Z	U	V	Z
<i>Positive correlation between faithfulness and row margins</i>									
U	0.248	− 0.181	− 0.067	0.248	− 0.181	− 0.067	0.225	− 0.150	− 0.075
V	− 0.182	0.251	− 0.069	− 0.182	0.251	− 0.069	− 0.149	0.228	− 0.079
Z	− 0.066	− 0.069	0.136	− 0.066	− 0.069	0.136	− 0.075	− 0.078	0.153
<i>Negative correlation between faithfulness and row margins</i>									
U	− 0.270	0.206	0.064	− 0.270	0.206	0.064	− 0.268	0.204	0.064
V	0.205	− 0.271	0.066	0.205	− 0.271	0.066	0.202	− 0.269	0.066
Z	0.066	0.065	− 0.131	0.066	0.065	− 0.131	0.066	0.065	− 0.131

G = Goodman, K = King OLS, BP = Brown and Payne

and assume that employment status at the individual level affects the probability that a voter chooses one of the two available options in the first election. This induces correlation between the party chosen in the first election and the proportion of unemployed. In the second election we assume that voter decisions depend both on the party voted in the previous election and on whether the voter is employed or not.

The results in Table 2 indicate that, when covariates are not taken into account, all three method are heavily biased in the same direction. Because, due to the large size of the data set, we may expect that errors due to random variation should not exceed 0.005, we may conclude that the estimates provided by both the King and the Bron and Payne methods with covariates are, again, both heavily biased, though in a different way.

4.3 Detecting ecological bias by bootstrap

The two first data sets described above were also used to test the following statements:

Table 2 True and ecological estimates of voter transitions in example 2

Method	T		KC		BC	
	U	V	U	V	U	V
U	0.2583	0.7417	0.0461	0.9539	0.1399	0.8601
V	0.1462	0.8538	0.3588	0.6412	0.2648	0.7352
Method	G		K		B	
	U	V	U	V	U	V
U	0.4042	0.5958	0.4104	0.5896	0.4043	0.5957
V	0.0001	0.9999	0.0000	1.0000	0.0000	1.0000

T = true, K = King, B = Brown–Payne and G = Goodman; C stands for model with covariates

1. If a model that does not account for the effect of covariates is fitted, the bootstrap standard errors of transition probabilities are likely to be larger than those expected under a correctly specified model.
2. If a correctly specified model is fitted, the bootstrap and the theoretical standard errors tend to be of approximately the same size.

Let N be the number of times that the standard error estimated by bootstrap is larger than the corresponding one estimated under ideal conditions; let also p be the probability that a symmetric binomial distribution with RC trials is not smaller than N . Statement 1 will be accepted if p does not exceeds the significance level 0.025, meaning that N is large and close to RC . Statement 2 will be accepted if both $1 - p$ and p exceed 0.025, that is, N is neither too large, nor too small.

To avoid applying directly a bootstrap to the 20,000 polling stations available in the two data sets, which would require a huge computing time, 400 samples of 900 polling stations each were selected at random with replacement from each data set; this is approximately the number of polling stations in the real data set of Sect. 5. The results are summarize in Table 3 which reports also the results obtained when correlations are null; these results indicate clearly that both statements above can be accepted.

5 An application to real data

This study concerns the second ballot in the 2016 Municipal elections in Turin, one of the largest urban area in Northern Italy, Though in the Italian system, since 1993, Municipal elections are based on a two-round system, during the two most recent elections in Turin, held in 2006 and 2011, the candidate of the centre-left coalition was elected in the first round. Considering the traditional strength of the centre-left coalition in Turin and that, due to personal rivalries, the parties on the centre-right were running with three different candidates, most political analysts had predicted that candidate Piero Fassino, incumbent mayor and one of the Democratic Party (PD) leader at the national level, would probably win in the first ballot.

Since 2013 the Italian political scenario is considered to be a tripolar party system, with the Movimento 5 Stelle (M5S) having gained a strength comparable to that of the more traditional centre-left and centre-right coalitions, a feature which has, gradually, extended to several local political systems; see, for example, Emanuele and Chiaramonte (2013) and Regalia (2018). Though the M5S was active in Turin since the 2011 Municipal election, only after the 2016 Municipal election M5S became the third-largest political group in the local scenario; the M5S candidate was Chiara Appendino, former party chief councillor.

Table 3 Bootstrap results: number of times that the standard error computed from theory is not larger than the one computed from bootstrap and the corresponding p value

	Positive correlation		Negative correlation		Null correlation
	C	NC	C	NC	
N	5	9	4	9	5
$\min(p, 1 - p)$	0.500	0.002	0.500	0.002	0.500

C = model with covariates, NC = model without covariates

On the whole there were six main candidates running at the first ballot; the remaining ones were grouped into two residual categories:

1. Airaudo, (LEFT), heading a left-wing post communist coalition;
2. Fassino, (PD), heading a centre-left coalition;
3. Appendino, (M5S), candidate of the Five-Stars Movement;
4. Napoli, (FI), running for Forza Italia, Berlusconi's party;
5. Morano, (LN), Lega Nord's candidate;
6. Rosso (UDC), another centre-right party;
7. (ODX), other candidates on the right;
8. (OSX), other candidates on the left;
9. (NOV), in both ballots this category denotes abstainer, blank votes and other unclassified votes

The 2016 first-round election took place on June 5th 2016 and, surprisingly, the centre-left leader did not win as expected and Fassino (PD) and Appendino (M5S) entered into a runoff election which was held 2 weeks later. Though the expected winner was again the centre-left coalition leader, he was clearly defeated, a result that represents a true turning point in the Turin local political history. The objective of the estimation and analysis of voter transitions from the first to the second ballot tries to throw some new light on what happened. One conjecture that we aim to verify is that a substantial amount of voters of centre-right candidates excluded from the ballot voted M5S rather than abstaining, which would be an instance of strategic voting (Herrmann and Pappi 2008), when voters make the effort to go to the ballots to prevent the victory of the candidate they disliked most.

Turin has 909 valid polling stations, after removing hospitals, prisons and the like, with an average of 786 voters each. To account for possible heterogeneity in voting behaviour, we used a geographic information system software (GIS) to determine the position of the 909 polling stations and link to the most relevant social-economic variables from Census data. By matching polling stations with census data, the following five covariates, measured at the level of polling stations and expressed as ratios with respect to the number of eligible voters, were made available: aged 65 and over, registered immigrants, unemployed, low qualified workers and people with low education. A map of Turin macro areas is given in Fig. 2 and the average value of covariates for macro areas are plotted in Fig. 3. We started by fitting a modified Brown and Payne model without covariates and then tried several alternative models which allow the most relevant cells of the table of voter transitions to depend on the available covariates. An informal model selection procedure was used by comparing the estimates of the various regression coefficients on the logit scale with the corresponding standard errors and removing non-significant covariates. The fact that the final model with covariates is a substantial improvement is supported by the substantial increase in the log-likelihood from about $-11,800$ to -5680 and the fact that all estimated regression coefficients are significant at the 5% level. Estimates of transitions provided by the model without covariates and by the final model are displayed in Table 4.

Estimates of the regression coefficients and the corresponding ratios z between estimates and standard errors for the model with covariates are displayed in Table 5. The absolute value of the z ratio gives a measure of significance in the sense that, the greater the value, the more unlikely it is that the corresponding association is a random artifact; usually values greater than 1.96 are taken as significant. In addition, the log-likelihood increases from $-11,848$ to -5682 .

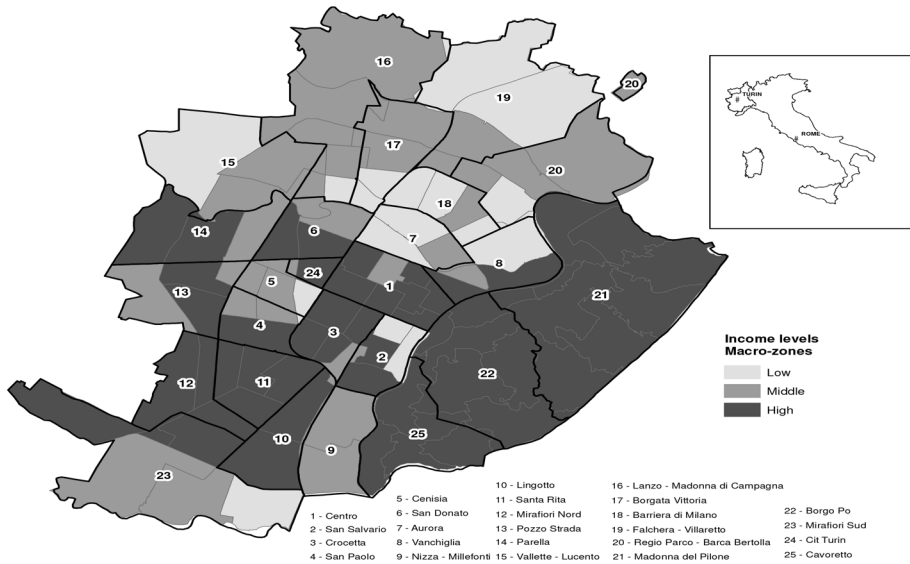


Fig. 2 Map of zones in Turin borough, darker area correspond to higher average income

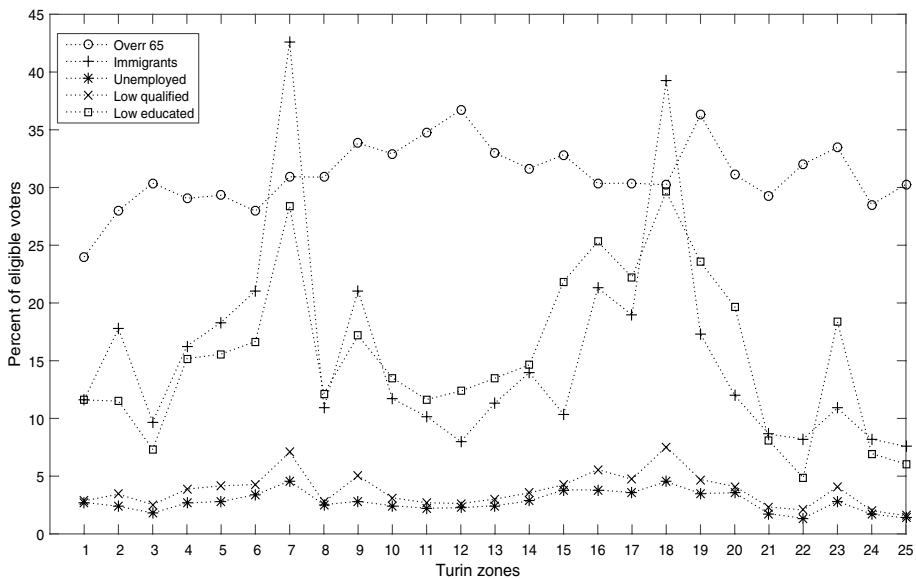


Fig. 3 Average covariate values within the 25 macro areas composing the Municipality of Turin

Table 5 lists the z ratios for entries of the table of voter transitions where a covariate was used. When the value of z in a given cell is positive and the corresponding covariate increases, the estimated transition also increases, keeping all other covariates fixed. For instance, when the proportion of aged people (or that of immigrants) increases, the entry in cell (1,2) also increases; instead, it decreases when the proportion of low

Table 4 Estimates of voter transitions between the first and the second ballot in the city of Turin with and without covariates

	First ballot	Second ballot					
		Without covariates			With covariates		
		M5S	PD	NOV	M5S	PD	NOV
LEFT		0.000	0.894	0.106	0.093	0.841	0.066
M5S		1.000	0.000	0.000	1.000	0.000	0.000
UDC		1.000	0.000	0.000	0.947	0.000	0.053
FI		0.828	0.096	0.076	0.797	0.203	0.000
PD		0.033	0.944	0.023	0.036	0.902	0.061
LN		0.808	0.096	0.096	0.754	0.153	0.092
ODX		1.000	0.000	0.000	0.479	0.000	0.521
OSX		0.496	0.000	0.504	0.600	0.189	0.211
NOV		0.005	0.000	0.995	0.025	0.006	0.969

Table 5 Estimated regression coefficient for the model with covariates

Cov	Row	Col	Est.	<i>z</i>	Cov	Row	Col	Est.	<i>z</i>
An	1	2	0.59	3.20	Bq	4	1	− 0.85	− 2.98
Im	1	2	0.44	3.21	An	5	2	− 0.01	− 2.08
Bs	1	2	− 0.96	− 3.19	Bs	6	1	0.16	4.63
An	4	1	− 0.14	− 4.61	Di	6	1	− 0.50	− 3.92
Im	4	1	− 0.17	− 3.09	Im	7	1	− 0.21	− 4.13
Bs	4	1	0.92	5.27	Bq	7	1	0.42	2.37

An = proportion of voters ages 65 and over; Im = proportion of immigrants; Bs = proportion of residents with low schooling, Bq = proportion of low qualified workers; Di = proportion of unemployed

Table 6 Comparison between standard error of estimated voter transitions based on bootstrap and on theory

Without covariates		With covariates	
<i>N</i>	<i>p</i>	<i>N</i>	<i>p</i>
23	0.0002	18	0.0610

N = number of times that the bootstrap standard error is larger than the one based on theory; *p* is the significance level

educated people increases. To gain further insights, the final model was used to compute estimated vote transitions within four macro zones, two belonging to the central area (1,3) and two in the surroundings (15, 23). For instance, voters from the LEFT that voted Fassino is above 82% in the central area against less than 63% in the surroundings; also, the proportion of voters from UDC who voted Fassino is above 37% in the central area against less than 3% in the surroundings.

Additional evidence that the estimates of voter transitions with covariates gives a closer approximation to the truth is provided by the comparison of the estimates of the standard errors of voter transitions according to theory (assuming no ecological bias) and by bootstrap, for the models with and without covariates. Formally, the results

displayed in Table 6 indicate that the assumption of equivalence between standard errors based on bootstrap and on theory is clearly rejected when covariates are not used. Though in the model with covariates the bootstrap estimates are larger than the theoretical ones 18 times, this is not significant; formally, the excess of positive differences is due to random variation, but a slight amount of bias may still be present.

6 Concluding remarks

One of the main conclusions of this paper is that, whenever the variations in the proportion of voter transitions across polling stations are correlated with the strength of certain parties in the previous election, estimates from aggregated data will be biased, no matter which method of ecological inference is used. Though methods of ecological inference differ in the way they model the randomness intrinsic to voters' decisions, when the number of available polling stations is relatively large, this component may be almost negligible relative to the potential ecological biases.

It is important to emphasize that, by modelling the effect of aggregate level covariates on transition probability, we are implicitly assuming that they act as *macro properties* (Firebaugh 1978), meaning that certain average features of polling stations affect the decisions of those who voted in the same way in the previous election; a detailed study of situations where this may happen are discussed by Johnston and Pattie (2006). However, there may be situations where voters' decisions depend on the individual level of certain covariates (e.g. age, employment condition, income) rather than on the corresponding values aggregated within polling stations. When this is the case, the model based on aggregate level covariates (either King or Brown and Payne) constitute an incorrectly specified model which may fail to remove ecological bias. We describe a data generating mechanism of this type leading to substantial bias which cannot be removed by modelling covariates.

It is well known that, unless data at the individual level are available, it will be difficult to find objective evidence of ecological fallacy. However, when aggregate level covariates are available, indirect evidence of bias may be achieved by fitting models with covariates. To be specific, when fitting a model which allows transition probabilities to depend on a given covariates, two related statistical checks may be performed: (1) does the model fit significantly better relative to the one without covariates? (2) are the estimated regression coefficients highly significant? When both these conditions are satisfied, it is very unlikely that the detected dependence of transitions from covariates is a numerical artifact.

An additional technique which we propose for detecting ecological bias, is based on comparing the standard errors of the estimated transitions obtained from bootstrap with those based on theory; we describe a simple testing procedure based on the symmetric binomial distribution. The procedure is applied successfully to two artificial data sets with and without bias and in the application to the Turin election. However one has to keep in mind that this is a computationally demanding procedure which require additional scrutiny when estimated transitions are either close to 0 or 1.

Concerning the linear regression method of ecological inference, also known as the Goodman method, our investigation indicates that its main limitation is that it cannot account for the effect of covariates properly. Goodman (1959) mentioned the possibility to model the effect of covariates, but Chambers and Steel (2001) show that such an approach may be problematic. Because of this, it should be applied only within an area which is sufficiently small to believe that transition probabilities do not differ in a systematic way

across polling stations. For instance, the results presented in the previous section suggest that, though the estimate provided by Goodman's method are likely to be partly biased when applied to the city of Turin as a whole, its application should probably be valid within suitably chosen sub-areas of the Municipality. In doing so, we need to balance two conflicting requirements: (1) the number of polling stations should not be too small, otherwise estimates will be affected by large random fluctuations; (2) if the area is too large, it is likely that estimates are affected by a certain amount of ecological bias. An advantage of the Goodman's method is that a bootstrap estimate of the standard errors of voter transitions is much easier to compute than in any other methods.

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